Yifan Yang

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RESEARCH INTERESTS

- Spatiotemporal Complementarity
- Place Specificity vs. Transferability
- Accountability in GeoAI
- Responsible and Agentic GeoAI
- Generative AI and Multimodal Learning
- Street-View Imagery and Remote Sensing
- Disaster Damage Assessment and Resilience
- GIScience and AI for Science (AI4Science)

Vision: To transcend the boundaries of screens and make the real world our playground of intelligence, where AI, space, and humanity coexist and co-create.

EDUCATION

2024-pres. **Ph.D.** (expected 2027), Geography (GIScience), Texas A&M University (TAMU), College Station, TX
GPA 3.80/4.0. Advisor: Dr. Lei Zou (Geospatial Exploration and Resolution (GEAR) Lab). Committee: Dr. Heng Cai, Dr. Andrew Klein, and Dr. Zhengzhong Tu. Passed the preliminary examination on June 24, 2026, and the dissertation proposal defense on November 13, 2026 (Ph.D. candidate).

2022-2024 **M.S.**, Spatial Data Science, University of Southern California (USC), Los Angeles, CA
GPA 3.96/4.0. Advisors: Dr. John P. Wilson and Dr. Siqin Wang

2018-2022 **B.E.**, Software Engineering, Hainan University, Haikou, China
Average score 87.4/100. Advisors: Dr. Jieren Cheng and Dr. Qi Qi

APPOINTMENTS

- **Academic Institutions**

2025-pres. Graduate Research Assistant, GEAR Lab, Department of Geography, Texas A&M University (TAMU). Advisor: Dr. Lei Zou

2026 Student Researcher, NSF I-GUIDE Summer School 2026: Spatial AI & Convergence Science, University of Illinois Urbana-Champaign (July 13-17, 2026)

2026 Student Researcher, Institute for a Disaster Resilient Texas (IDRT), TAMU (Apr-Aug 2026). Advisor: Dr. Jooho Kim

2025 Research Intern, Spatial Data Lab, Center for Geographic Analysis, Harvard University (Remote; Mar-Aug 2025). Mentor: Dr. Siqin Wang

2024 Graduate Teaching Assistant, Department of Geography, TAMU (Fall 2024)

2023-2024 Master Student Researcher, Urban Tree Shade Model, Spatial Sciences Institute, USC (Sep 2023-Apr 2024). Advisors: Dr. John P. Wilson, Dr. Yi Qi, and Beau MacDonald

- **Industry**

  2023 Data Engineering and AI Model Fine-tuning Intern, Vitally AI (Remote; Feb-May 2023)

- **Professional Organizations**

  2026-2027 Student Director, Geographic Information Science and Systems (GISS) Specialty Group, American Association of Geographers (AAG)

  2025-2027 Student Co-Director, Remote Sensing Specialty Group, AAG

  2025-2027 Student Co-Director, Hazards, Risks, and Disasters Specialty Group, AAG

  2026-2027 Communications Fellow, GeoSAT Student Fellowship Program

  2025-pres. Social Media Outreach Representative, *Annals of GIS*

  2024-2025 Board Member, GISphere (lead of the GISalon Roundtable Series)

  2021-2022 Vice President, Robotics and Artificial Intelligence Association, Hainan University

HONORS AND AWARDS

2026 Travel Award, 34th ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems (SIGSPATIAL 2026), Riverside, CA, $500

2026 2nd Place, AAG GIS Specialty Group (GISSG) Student Honors Paper Competition, $350

2026 AAG International Geographic Information Fund (IGIF) Scholarship Award, $1,200

2026 Doctoral Scholarship Award, Cartography and Geographic Information Society (CaGIS), $1,000

2026 Geography Graduate Student Enhancement Scholarship, Department of Geography, TAMU, $1,111

2026 GEOG Graduate Travel Funding (2026-2027), Department of Geography, TAMU, $1,000

2026 Communications Fellow, GeoSAT Student Fellowship Program (2026-2027), $500

2026 Selected Travel Award, NSF I-GUIDE Summer School: Spatial AI & Convergence Science, University of Illinois Urbana-Champaign, over $2,000

2026 IEEE GRSS Travel Grant, IGARSS 2026, Washington, D.C., $500

2025 Best Student Paper Award, International Cartographic Conference (ICC 2025), Vancouver, Canada

2025 Honorable Mention (Top 5), AAG GISSG Student Honors Paper Competition, $200

2025 Environment and Sustainability Graduate Fellow Award, TAMU, $2,500

2025 Student Ambassador Scholarship (Domain Data Science Track), Texas A&M Institute of Data Science (TAMIDS), $2,000

2025 Selected Abstract, NSF I-GUIDE Spatial AI Challenge: GeoAgent4Disaster, an autonomous multi-agent framework for multimodal disaster assessment (selected two consecutive years)

2025 CaGIS International Travel Grant, ICC 2025, Vancouver, Canada, $2,100

2025 GEOG Graduate Travel Funding (2025-2026), Department of Geography, TAMU, $1,100

2025 Annual Meeting Award, AAG Applied Geography Specialty Group (AGSG), $196

2025 Travel Grant, Summer School on Cyberinfrastructure and Disaster Resilience (NSF-funded), TAMU, $200

2024 Most Suitably Applied Analysis Methodology, ArcGIS StoryMaps Competition, Los Angeles Geospatial Summit

2024 Selected Abstract, NSF I-GUIDE Spatial AI Challenge: Perceiving Multidimensional Disaster Damages from Street-View Images Using Visual-Language Models
2024 Lifetime Membership, Nu Theta Chapter, Gamma Theta Upsilon (International Geographic Honor Society)
2024 GEOG Graduate Travel Funding (2024-2025), Department of Geography, TAMU, $1,000
2024 Visiting Travel Grant, Department of Geography, University of South Carolina, $400
2023 Gold Award, 9th China International "Internet+" College Students Innovation and Entrepreneurship Competition, Higher Education Main Track, International Project Special Competition (Shanghai Division), for "Ushine: AI+ Multilingual Talent Service Platform"
2020-2021 First-Class Comprehensive Scholarship, Hainan University, ¥2,500
2020-2021 "College Student with the Most Innovative Spirit and Practical Ability," Hainan University
2020 National First Prize, Computer Design Competition for Chinese College Students (Big Data Practice)
2020 Silver Award (Hainan Creative Group), 6th China International "Internet+" Student Innovation and Entrepreneurship Competition
2020 Second Prize, 3rd China Youth Cup National University Student Mathematical Modeling Competition
2020 Second Prize, 14th iCAN International Innovation and Entrepreneurship Competition (South China Region)
2020 Second Prize, 13th "Certification Cup" Mathematics China Mathematical Modeling Network Challenge (Inner Mongolia Region)
2020 Third Prize, 10th MathorCup College Mathematical Modeling Challenge
2020 Third Prize, Innovation Group (Artificial Intelligence Track), 2020 China University Computer Competition
2020 Third Prize, China-US Youth Creators Competition (Haikou Region)
2020 Third Prize, 6th National Mobile Internet Innovation Competition (South China Region)
2020 Third Prize (Finalist), Hainan Selection Competition of the 4th China Creative Wings Innovation Competition
2020 Bronze Prize (Hainan Region), Challenge Cup Student Entrepreneurship Plan Competition
2020 Honorable Mention, 2020 Hainan Free Trade Port Entrepreneurship Competition

RESEARCH PROJECTS

- **Selected Research Projects (Lead Developer)**

2026-pres. **RAY (Resilience Assistant for You): An Accountable GeoAI Harness for Place-Based Disaster Intelligence.** An evidence console (WebGIS / smartphone PWA) in which an open-model analyst plans its own spatial analyses, while a Steward Harness authorizes every step, checks every number against the result it cites, and replays any run without the model. Manuscript: "Accountability Enforced, Not Promised: A Harness for GeoAI Agents in Disaster Intelligence." [GitHub] [Live app] [Video]

- Built four public-data cases (Eaton and Palisades fires 2025, Hurricanes Milton 2024 and Ian 2022) on H3 tiles, with a resident mode (tile-level dossier, never a parcel

statement) and a planner mode (inspection priority from four weighted criteria, published presets, hashed source lineage)
- Authorization: a proof plus a model-free study over 2,115 enumerated plans (5,309 steps): 0 of 2,056 authorized steps leak, against 437 leaks for a check that runs dependents of denied steps and 465 for an output filter
- Numeric provenance: the citation check refuses 91% of 10,440 injected errors and 0 of 1,347 truth-preserving rewrites in 936 answers from four open models
- 123-task analyst benchmark across four open models: governed runs succeed on 88% of tasks with 0 denied steps run; without the plan check, 77% and 25 denied steps run; further studies on seven open vision models, map reading, burn severity, and clarifying questions

2026-pres. **PrepStreet: Seeing Hurricane Damage Before It Happens.** Multimodal GeoAI that predicts street-level hurricane damage from information available before landfall (pre-event street views, 10-m AlphaEarth embeddings, NHC forecasts, geospatial context), generates severity-conditioned post-storm street scenes, and integrates both into risk maps and inspection rankings benchmarked against FEMA flood zones. Extended abstract, AAG 2027 Spatial Analysis and Modeling Specialty Group (John Odland Award). [GitHub] [Project page]
- Built leakage-audited benchmarks grouped by street segment or block on Hurricane Milton (2,556 co-located pre/post panoramas, 40 segments) and Hurricane Ian (4,078 Mapillary samples, 166 blocks)
- Pre-event information recovers 93% of a post-disaster image model's macro-F1 on Milton (0.599 vs. 0.646) and 86-93% on Ian; AlphaEarth embeddings and street views are complementary across damage types
- Learned rankings reach Spearman $\rho = 0.79$ (Milton) and 0.83 (Ian), while FEMA flood zones rank damage in reverse ($\rho = -0.27$) or more weakly ($\rho = 0.47$)
- Severity-conditioned generation preserves the depicted street (LPIPS 0.37-0.39, SSIM 0.46-0.47) and grows more faithful to observed damage as the condition improves (0.31 → 0.40 → 0.48)

2025-2026 **Sat2Street-DisasterGen: Satellite-to-Street View Synthesis for Post-Disaster Assessment.** A multimodal generative framework that benchmarks cross-view image generation strategies integrating diffusion models, ControlNet, and VLM-guided semantic conditioning to synthesize realistic street-view imagery from post-disaster satellite imagery. [GitHub]
- Designed a unified evaluation framework with consistent satellite inputs, data splits, and protocols across generative methods
- Implemented cross-view generation pipelines: Pix2Pix (cGAN), SD1.5 + ControlNet, VLM-guided synthesis, and Disaster-MoE (Mixture-of-Experts)
- Built a multi-stage LoRA training pipeline for damage-severity-aware generation (Mild / Moderate / Severe)
- Evaluated outputs with pixel-level metrics (FID, SSIM, LPIPS), semantic consistency (ResNet-based classification), and VLM perceptual scoring

2025-2026 **RAPID: A Reproducible Multi-Agent Pipeline for Interpretable Disaster Damage Assessment from Satellite and Street-View Imagery.** An autonomous multi-agent GeoAI framework for hyperlocal, interpretable, near-real-time damage assessment that orchestrates specialized agents built on vision-language foundation models for cross-view

perception, damage recognition, reasoning, and automated report generation without task-specific retraining. [GitHub]

- Designed a modular multi-agent architecture: Disaster Perception, Image Restoration, Damage Recognition, and Disaster Reasoning agents
- Integrated satellite imagery, street-view imagery, textual cues, and temporal change signals; zero-/few-shot analysis with VLMs (e.g., GPT, Gemini)
- Enabled cross-view semantic alignment and causal reasoning for damage assessment and recovery recommendations, with an interpretable reporting pipeline

2025-2026 **DamageArbiter: CLIP-Enhanced Multimodal Framework for Hurricane Damage Assessment.** A multimodal arbitration framework for hyperlocal hurricane damage assessment from bi-temporal street-view imagery, integrating vision-language models (CLIP) with deep vision backbones (ViT-B/16) to improve robustness under post-disaster uncertainty. [GitHub]

- Designed a multimodal arbitration mechanism to reconcile conflicting visual and semantic signals
- Implemented ViT-B/16 and transformer-based classification pipelines; evaluated with precision, recall, F1, and cross-validation on hurricane street-view datasets

2025-2026 **BiTemporal-StreetView-Damage: Hyperlocal Disaster Assessment via Bi-Temporal Street-View Imagery.** A dual-channel framework using pre-trained vision backbones (Swin Transformer, ConvNeXt) on pre- and post-disaster street-view pairs to improve the accuracy and interpretability of damage-severity estimation. [GitHub]

- Constructed a dataset of 2,249 labeled street-view image pairs with fine-grained severity annotations
- Improved classification accuracy from 66.14% (post-only) to 77.11% (bi-temporal); evaluated cross-attention, weighted fusion, and Siamese-difference strategies
- Applied Grad-CAM to demonstrate enhanced spatial attention from pre-disaster inputs, enabling fine-grained damage mapping for climate-resilient planning
- **Industry engagement**: featured by Mosaic - Geospatial Imaging Leaders (Prague, Czech Republic) for 360° street-level surveying and disaster-intelligence applications [link]

- **Open-Source Community Leadership**

2026-pres. **Founder**, AutonomousGeoAI4Science (AutoGeoAI4Sci), an open research community and hub for papers, tools, benchmarks, and collaboration advancing autonomous and multimodal GeoAI for scientific discovery. [GitHub Organization]

- Curate and maintain 60+ repositories on cross-view GeoAI, disaster intelligence, and autonomous spatial reasoning
- Develop open-source tutorials and research frameworks for multimodal geospatial AI; promote reproducible research connecting AI, GIScience, and Earth observation communities

PUBLICATIONS

Notes:

- **Yang, Y.** denotes myself
- Google Scholar: 443 citations, h-index 8, i10-index 7 (as of November 2026)
- **Total**: 33 (published/accepted: 25; in review: 3; preprints: 5)
  - + First- or Corresponding-Authored Publications: 11 (journal articles: 3, conference papers: 5, book chapters: 2, preprints: 1)
  - + Co-Authored Publications: 22 (journal articles: 11, conference papers: 7, preprints: 4)

- **First- or Corresponding-Authored Publications (in chronological order)**

Review 11. **Yang, Y.**, Zou, L., Gong, W., Fu, K., Li, Z., Wang, S., Zhou, B., Cai, H., Tian, H. DamageArbiter: A Multimodal Arbitration Framework for Disaster Damage Assessment from Street-View Imagery. *Natural Hazards*. [arXiv:2603.14837] **2nd Place, AAG GISSG Student Honors Paper Competition 2026**

Review 10. **Yang, Y.**, Zou, L., Tian, H., Zhou, B., Abedin, J., Li, Z., Tu, Z. Generative AI for Disaster Resilience. In G. Mai, X. Huang, D. Jain, & D. Lunga (Eds.), *Geography in the Age of Generative AI: Innovations in Mapping, Analysis & Geospatial Applications*. CRC Press / Taylor & Francis. [project page]

2026 9. Chen, J., **Yang, Y.** (corresponding author) MERIDIAN: Memory-Encoded Rituals for Intimacy across Distance in an Interactive, Asynchronous Neighbourhood. *arXiv preprint*. [arXiv:2610.41776]

2026 8. **Yang, Y.**, Zou, L. When Satellite and Street Views Disagree: Cross-View Complementarity in Disaster Damage Assessment. *Proceedings of the 5th ACM SIGSPATIAL International Workshop on Searching and Mining Large Collections of Geospatial Data (GeoSearch '26)*, Riverside, CA (Lightning talk, 4 pages; Accepted). [arXiv:2610.04327]

2026 7. **Yang, Y.**, Gong, W., Zhang, K., Zou, L., Tu, Z., Li, H., Li, Z., Ye, X. RAPID: A Reproducible Multi-Agent Pipeline for Interpretable Disaster Damage Assessment from Satellite and Street-View Imagery. *Proceedings of the 34th ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems (SIGSPATIAL '26)*, Riverside, CA (Applications Track full paper, 12 pages; Accepted). [link]

2026 6. **Yang, Y.**, Zou, L. RAPIDMap: Rapid Multi-Agent Pipeline for Interpretable Disaster Mapping from Satellite and Street-View Imagery. *Cartography and Geographic Information Society (CaGIS) Conference 2026*, St. Louis, MO (Accepted). [link]

2026 5. **Yang, Y.**, Zou, L., Jepson, W. Satellite-to-Street: Synthesizing Post-Disaster Views from Satellite Imagery via Generative Vision Models. *2026 IEEE International Geoscience and Remote Sensing Symposium (IGARSS 2026)*, Washington, D.C., 1764-1768. [link]

2025 4. **Yang, Y.**, Zou, L., Zhou, B., Li, D., Lin, B., Abedin, J., Yang, M. Hyperlocal Disaster Damage Assessment Using Bi-Temporal Street-View Imagery and Pre-Trained Vision Models. *Computers, Environment and Urban Systems*, 121, 102335. [link] **Honorable Mention, AAG GISSG Student Honors Paper Competition 2025**

2025 3. **Yang, Y.**, Zou, L. Perceiving Multidimensional Disaster Damages from Street-View Images Using Visual-Language Models. *Abstracts of the International Cartographic Association*, 10, 310. [link] **Best Student Paper Award, ICC 2025**

2025 2. **Yang, Y.**, Borrelli, D. Object Detection and Segmentation of Trees Using Text SAM in ArcGIS Online. In D. M. Ruddell & D. Ter-Ghazaryan (Eds.), *Security First: Geospatial Workflows for a Safe and Equitable World* (Chapter 7, pp. 113-122). Redlands, CA: Esri Press. [link]

2024 1. **Yang, Y.**, Wang, S., Li, D., Sun, S., Wu, Q. GeoLocator: A Location-Integrated Large Multimodal Model (LMM) for Inferring Geo-Privacy. *Applied Sciences*, 14(16), 7091. [link]

- **Co-Authored Publications: Journal Articles**

Review 11. Li, Z., Zou, L., Seo, J.H., Yang, M., **Yang, Y.** Seeing Green from Indoors in 3D: How Built Environment and Vegetation Shape Window-Level Nature Exposure. *Sustainable Cities and Society*. [SSRN:6766522]

2026 10. Tian, H., Cai, H., Chen, X., **Yang, Y.** Monitoring Urban Traffic Dynamics at High Spatiotemporal Resolution Using Distributed Acoustic Sensing and Deep Learning. *Journal of Transport Geography*, 138, 104901. [link]

2026 9. Lin, B., Zou, L., Tian, H., Cai, H., **Yang, Y.**, Zhou, B. Modeling Hospital Visitation Flows Through Distance-Attractiveness Tradeoffs: Integrating Human Mobility, Hospital Attributes, and Socioeconomic Context. *Transactions in GIS*, 30(7), e70432. [link]

2026 8. Kim, J., Shakya, A.R., Kelly, J., **Yang, Y.**, Lee, S., Brody, S.D., Mostafavi, A. Debris-Cast: A Two-Stage Machine Learning Framework for Hurricane Debris Prediction Using Multi-Source Geospatial Data. *International Journal of Disaster Risk Reduction*, 146, 106470. [link]

2026 7. Wang, S., Liu, H., Wu, C.Y., Huang, X., Wang, R., **Yang, Y.**, Corcoran, J., Lai, S., Xia, X., Liu, Y. The Boiling Frog Effect: Global Warming Delays Emotional Impacts of Air Pollution in Warmer Climates. *Journal of Hazardous Materials*, 142440. [link]

2026 6. Li, Z., Li, H., **Yang, Y.**, Wang, S., Zhu, Y. Integrating Earth Observation Data into the Tri-Environmental Evaluation of the Economic Cost of Natural Disasters: A Case Study of the 2025 LA Wildfire. *International Journal of Applied Earth Observation and Geoinformation*, 150, 105342. [link]

2026 5. Ye, X., Gong, W., **Yang, Y.**, Zou, L., Tu, Z., Huang, X., Li, Z., Ning, H., Wu, L. Towards Agentic Urban Digital Twins (AUDiTs): Advancing New Urban Science through Human-AI Co-Learning Agents. *Urban Informatics*, 5(1), 9.

2025 4. Wang, Y., Yin, H., **Yang, Y.**, Zhao, C., Wang, S. Navigating Spatial Inequities in Freight Truck Crash Severity via Counterfactual Inference in Los Angeles. *Journal of Transport Geography*, 128, 104387.

2025 3. Gong, W., Ye, X., Wu, K., Jamonnak, S., Zhang, W., **Yang, Y.**, Huang, X. Integrating Spatiotemporal Vision Transformer into Digital Twins for High-Resolution Heat Stress Forecasting in Campus Environments. *Journal of Planning Education and Research*, 1-15. [link]

2020 2. Zhang, Y., Chen, J., **Yang, Y.**, Li, H., Zheng, X., Chen, X., Liu, B., Ren, T., Xiong, N. COVID-19 Public Opinion and Emotion Monitoring System Based on Time Series Thermal New Word Mining. *Computers, Materials & Continua*, 64(3), 1415-1434. [link]

2020 1. Shi, C., Mou, G., Wang, H., Liu, T., **Yang, Y.**, Li, Z. Different Loss Functions Used in the Low-Rank Approximation. *International Core Journal of Engineering*, 6(11), 360-368.

- **Co-Authored Publications: Conference Proceedings**

2026 7. Chen, T., **Yang, Y.**, Zou, L., Brannstrom, C., Harsuko, R., et al. Engaging Communities, Energizing Futures: A Framework for Geothermal Awareness and Acceptance. *AGU 2026 Fall Meeting*, San Francisco, CA (12/07-12/11, 2026; Accepted).

2026 6. Shakya, A., Kim, J., Kelly, J., **Yang, Y.**, Brody, S.D. Debris-Cast: Data-Driven Hurricane Debris Estimation for Early Recovery Planning. *AGU 2026 Fall Meeting*, San Francisco, CA (12/07-12/11, 2026; Accepted).

2026 5. Kim, J., **Yang, Y.**, Shakya, A., Kelly, J. Optimizing HPC Performance for CDC-Based Storage and Reconstruction of High-Resolution Imagery. *SC26: The International Conference for High Performance Computing, Networking, Storage, and Analysis*, Chicago, IL (Research Poster; Accepted). [link]

2026 4. Zou, L., Zhou, B., **Yang, Y.** Responsible GeoAI: Thoughtful Integration of GIScience and AI in the Real World. *1st International Conference on Geospatial Artificial Intelligence (GeoAI 2026)*, Ghent, Belgium.

2026 3. Tian, H., Cai, H., **Yang, Y.** BEACON: Behavioral and Semantic Enrichment of AlphaEarth Embeddings through Tri-Modal Contrastive Learning. *Proceedings of the 34th ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems (SIGSPATIAL '26)*, Riverside, CA (Short paper; Accepted). [arXiv:2608.29553]

2025 2. Zou, L., Mandal, D., Zhou, B., **Yang, Y.**, Yang, M. Resilience in 4D: AI-Driven Geospatial Digital Twins for Urban Flood Simulation and Management. *AGU Annual Meeting 2025 (AGU25)*.

2021 1. Yan, B., Wang, J., Cheng, J., Zhou, Y., Zhang, Y., **Yang, Y.**, Liu, L., Zhao, H., Wang, C., Liu, B. Experiments of Federated Learning for COVID-19 Chest X-ray Images. *International Conference on Artificial Intelligence and Security (ICAIS 2021)*, pp. 41-53. Springer, Cham. [link]

- **Co-Authored Publications: Preprints**

2026 4. Li, Z., Yang, M., Zou, L., Ma, H., Tian, H., Zhou, S., et al., **Yang, Y.** GIScholarBench: Benchmarking LLM Overconfidence in GIS Research. *arXiv preprint*. [arXiv:2606.08036]

2026 3. Arora, A., Deng, J., Arora, A., Cai, H., Zou, L., **Yang, Y.** Does FIFA World Cup Performance Influence Tourism Arrivals?: Evidence from Cabo Verde and Other Countries. *SSRN preprint*. [SSRN:7596480]

2026 2. Gong, W., Srivastava, U., Wang, Y., Jia, Y., Wu, Q., Bai, W., **Yang, Y.**, Huang, X., et al., Ye, X. Earth Embeddings Reveal Diverse Urban Signals from Space. *arXiv preprint*. [arXiv:2604.03456]

2024 1. Wu, Q., Xu, Y., Xiao, T., Xiao, Y., Li, Y., Wang, T., et al., **Yang, Y.** Surveying Attitudinal Alignment between Large Language Models vs. Humans towards 17 Sustainable Development Goals. *arXiv preprint*. [arXiv:2404.13885]

TEACHING EXPERIENCE

- **Teaching Assistant**

Year	Term	Course	Institution
2024	Fall	GEOG 232: Cartography (lab sections 501, 502, 505)	Texas A&M University

- **Guest Lecturer**

Year	Term	Lecture Title	School/Course
2026	Spring	Geographically Weighted Regression (GWR)	TAMU, GEOG 475: Advanced GIS
2026	Spring	Git, GitHub, and Personal Websites for GIS Researchers	TAMU, GEOG 476: GIS Practicum
2025	Fall	Spatial Query Lab: Interactive Spatial SQL and PostGIS	TAMU, GEOG 659: GeoDatabases

PRESENTATIONS

Notes:

- Presenting author for all entries
- Total: 19

- **Conference Presentations**

2027 19. **Yang, Y.**, Zou, L. Seeing Hurricane Damage Before It Happens: Multimodal GeoAI for Street-Level Prediction and Visualization. *AAG 2027 Annual Meeting*, New York, NY (02/08-02/12, 2027; Accepted).

2026 18. Kim, J., **Yang, Y.**, Shakya, A., Kelly, J. Optimizing HPC Performance for CDC-Based Storage and Reconstruction of High-Resolution Imagery. Research Posters, *SC26*, McCormick Place, Chicago, IL (Poster; 11/17, 2026).

2026 17. **Yang, Y.**, Gong, W., Zhang, K., Zou, L., Tu, Z., Li, H., Li, Z., Ye, X. RAPID: A Reproducible Multi-Agent Pipeline for Interpretable Disaster Damage Assessment from Satellite and Street-View Imagery. Applications Track, Application 2: Disaster & Flood, *ACM SIGSPATIAL 2026*, Riverside, CA (11/04, 2026).

2026 16. **Yang, Y.**, Zou, L. When Satellite and Street Views Disagree: Cross-View Complementarity in Disaster Damage Assessment. Lightning Talk, *GeoSearch '26 Workshop at ACM SIGSPATIAL 2026*, Riverside, CA (11/03, 2026).

2026 15. **Yang, Y.**, Zou, L., Chen, T., Brannstrom, C., Chen, X., Harsuko, R., Nakata, N., Gross, M., van Wijk, J., Montazeri, S. GATES: A Geothermal Acceptance Tracking and Evaluation System Using Multi-Source Social Sensing. Data Science and Statistical Learning Session, *10th Annual Texas A&M Conference on Energy*, College Station, TX (Oral; 09/25, 2026).

2026 14. **Yang, Y.**, Fillman, K., Nawar, N., Thapaliya, R., Lyu, F. Detecting and Correcting Spatial Bias in VGI Using Remote Sensing. Invited Webinar Talk, *I-GUIDE Virtual Consulting Office (VCO)*, Online (09/23, 2026). [link]

2026 13. **Yang, Y.**, Zou, L. RAPIDMap: Rapid Multi-Agent Pipeline for Interpretable Disaster Mapping from Satellite and Street-View Imagery. Session 1B: GeoAI, Disaster Mapping, and Hydrography, *CaGIS Conference 2026*, St. Louis, MO (09/08-09/11, 2026).

2026 12. **Yang, Y.**, Zou, L., Jepson, W. Satellite-to-Street: Synthesizing Post-Disaster Views from Satellite Imagery via Generative Vision Models. Session: Leveraging AI, LLMs, and Geospatial Technologies for Rapid and Explainable Post-Disaster Damage Intelligence, *IEEE IGARSS 2026*, Washington, D.C. (08/13, 2026).

2026 11. **Yang, Y.** Satellite-to-Street: Synthesizing Post-Disaster Views from Satellite Imagery via Generative Vision Models. *Symposium on Spatiotemporal Data Science*, Virginia Tech, Alexandria, VA (Onsite Oral; 08/07-08/08, 2026).

2026 10. **Yang, Y.**, Zou, L. DamageArbiter: A Disagreement-Driven Arbitration Framework for Hurricane Damage Assessment from Street-View Imagery. GISS Specialty Group Honors Competition for Student Papers II, *AAG 2026 Annual Meeting*, San Francisco, CA (03/17-03/21, 2026).

2025 9. **Yang, Y.** Seeing Disagreements: An Explainable Multimodal Framework for Disaster Assessment and Spatial Mapping. Student Paper Competition, *Texas A&M University GIS Day (TxGIS Day 2025)*, College Station, TX (Oral; 11/17-11/21, 2025).

2025 8. **Yang, Y.**, Zou, L. Perceiving Multidimensional Disaster Damages from Street-View Images Using Visual-Language Models. Student Paper Session, *International Cartographic Conference (ICC 2025)*, Vancouver, Canada (08/18-08/22, 2025).

2025 7. **Yang, Y.**, Zou, L. DisasterVLP: A Vision-Language Pretrained Framework for Multidimensional Disaster Damage Assessment Using Street-View Images. *GeoAnalytics for Sustainable and Livable Cities*, ICC 2025 Pre-Conference Symposium, Vancouver, Canada (08/17, 2025).

2025 6. **Yang, Y.**, Gong, W., Zhang, K. DisasterVLP: Perceiving Multidimensional Disaster Damages from Street-View Images Using Visual-Language Models. *Summer School on Cyberinfrastructure and Disaster Resilience*, Texas A&M University, College Station, TX (Oral; 08/15, 2025).

2025 5. **Yang, Y.**, Zou, L. Hyperlocal Disaster Damage Assessment Using Bi-Temporal Street-View Imagery and Pre-Trained Image Processing Models. GISS Specialty Group Paper Competition II, *AAG 2025 Annual Meeting*, Detroit, MI (03/24-03/28, 2025).

2024 4. **Yang, Y.** GeoLocator: A Location-Integrated Large Multimodal Model (LMM) for Inferring Geo-Privacy. Thematic Session "Geoprivacy Challenges and Solutions in the Digital Society," *Spatial Data Science Symposium 2024*, Virtual. [video]

2024 3. **Yang, Y.** GeoLocator: A Location-Integrated Large Multimodal Model for Inferring Geo-Privacy. Symposium on GeoAI and Deep Learning for Geospatial Research: Human-Centered Geospatial Data Science III, *AAG 2024 Annual Meeting*, Honolulu, HI (04/16-04/20, 2024).

2024 2. **Yang, Y.** GeoLocator: A Location-Integrated Large Multimodal Model for Inferring Geo-Privacy. Multimodality Session, *AGI Leap Summit*, SuperAGI, Virtual (02/29, 2024).

2024 1. **Yang, Y.** GeoLocator: A Novel GeoAI Tool Making World Travel without Barriers & Urban Tree Shade Model. *2024 Los Angeles Geospatial Summit*, Los Angeles, CA (02/23, 2024).

ACADEMIC SERVICES

- **Conference Organizations**
  - 2027 Student Co-Lead, AAG Symposium on GIScience for Human Experience and Responsibility (GISER 2027), AAG 2027 Annual Meeting, New York, NY (02/08-02/12, 2027) [link]
  - 2027 Primary Organizer and Co-Chair, Panel "Responsible GeoAI: Challenges and Opportunities of Integrating Geography and AI in the Real World," AAG 2027 Annual Meeting, New York, NY (02/08-02/12, 2027; with Lei Zou and Yuhao Kang) [link]
  - 2027 Organizer and Chair, GeoAI Symposium: Panel Session on AI and Disaster Resilience, AAG 2027 Annual Meeting, New York, NY (02/08-02/12, 2027) [link]
  - 2026 Session Organizer, "Spatiotemporal AI for Urban and Environmental Intelligence: Sensing, Inference, and Human-Centered Applications," Symposium on Spatiotemporal Data Science, Virginia Tech, Alexandria, VA (08/07-08/08, 2026)
  - 2026 Session Co-Organizer, "GeoAI and Data Science for Disaster Resilience," 33rd International Conference on Geoinformatics (GeoInformatics 2026), Singapore (with Lei Zou, Yi Qiang, Yingjie Hu, Qunying Huang, Xiao Huang, and Bing Zhou) [link]
  - 2026 Session Co-Organizer, GeoAI and Deep Learning Symposium: GeoAI for Disaster Resilience, AAG 2026 Annual Meeting, San Francisco, CA (with Lei Zou, Yingjie Hu, Qunying Huang, Yi Qiang, Marcela Suarez, and Morteza Karimzadeh)
  - 2026 Organizer, Panel and Symposium, GISER Symposium: Past, Present, and Future of GIScience, AAG 2026 Annual Meeting, San Francisco, CA
  - 2026 Co-Organizer, AAG Remote Sensing Specialty Group Student Illustrated Paper Competition, AAG 2026 Annual Meeting, San Francisco, CA (with Ying Lu)
  - 2026 Program Committee Member, GeoSearch 2026: 5th ACM SIGSPATIAL International Workshop on Searching and Mining Large Collections of Geospatial Data (in conjunction with ACM SIGSPATIAL 2026, 11/03, 2026)
  - 2025 Session Co-Organizer, Symposium on Human Dynamics Research: Urban Environmental Intelligence and Human-Climate Interactions (Sessions I-II), AAG 2025 Annual Meeting, Detroit, MI (with Wenjing Gong, Sisi Wang, Xinyue Ye, and Xiao Huang)
- **University Service**
  - 2026 Organizer, TAMIDS Data Science Ambassador Seminar "From Signals to Society: Data Science for Understanding the Earth and Human Dynamics," Texas A&M University (03/2026)
  - 2025-2026 Data Science Student Ambassador, Texas A&M Institute of Data Science (TAMIDS)
  - 2025-2026 Executive Committee Member, Aggie Geographers Graduate Society (AGGS), Department of Geography, TAMU
- **Invited Reviews**

  ACM Transactions on Autonomous and Adaptive Systems
  Climatic Change
  Computational Urban Science
  International Journal of Applied Earth Observation and Geoinformation
  International Journal of Digital Earth
  International Journal of Geographical Information Science
  International Journal of ITS Research

International Journal of Remote Sensing
Journal of Transport Geography
npj Natural Hazards
Scientific Reports
Tourism Geographies
Transactions in GIS
32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD 2026), AI for Sciences Track
15th International Conference on Learning Representations (ICLR 2027)
2027 AAG GISS Specialty Group (GISS-SG) Student Honors Paper Competition

INDUSTRY ENGAGEMENT

2026-pres. Founder, Rayford AI: building a geospatial physical AI company that turns remote sensing, street-level imagery, and geospatial context into auditable intelligence for disaster assessment, claims triage, and infrastructure and property decisions [link]

2026 Invited Academic Exchange, Niantic Spatial, San Francisco, CA: discussion on GeoAI, multimodal spatial intelligence, and disaster-related spatial analytics, bridging academic research and large-scale spatial systems in industry (03/2026)

2026 Featured Research, Mosaic - Geospatial Imaging Leaders (Prague, Czech Republic): hyperlocal damage assessment using Mosaic street-level imagery from Hurricane Milton [link]

2023 Industry Internship, Vitally AI: data engineering and AI model fine-tuning for applied generative AI systems and production-oriented workflows

PROFESSIONAL SOCIETIES

2026-pres. ACM Special Interest Group on Spatial Information (ACM SIGSPATIAL)

2026-pres. IEEE Geoscience and Remote Sensing Society (IEEE GRSS)

2026-pres. International Association of Chinese Professionals in Geographic Information Sciences (CPGIS)

2025-pres. Cartography and Geographic Information Society (CaGIS)

2024-pres. American Association of Geographers (AAG)

2024-pres. Gamma Theta Upsilon (GTU), International Geographic Honor Society, Nu Theta Chapter (Lifetime Member)

TECHNICAL SKILLS

Programming: Python, C, C++, Java, SQL, R, JavaScript, HTML, ArcGIS API (JavaScript, Python)

Machine Learning: PyTorch, TensorFlow, Scikit-learn, HuggingFace Transformers and Diffusers, Stable Diffusion, LoRA/DreamBooth fine-tuning, vision-language models (CLIP, BLIP, GPT, Gemini)

Geospatial and Data Tools: Esri ArcGIS Pro, PostgreSQL & PostGIS, pgAdmin, MS SQL Server and SSMS, Git/GitHub, Jupyter, Anaconda, Docker, Linux/Unix, Visual Studio Code

SELECTED MEDIA COVERAGE

2026 Texas A&M College of Arts & Sciences, Five Texas A&M Researchers Join National Climate-Health Initiative [link]

2026 Mosaic - Geospatial Imaging Leaders, How Texas A&M Is Pioneering Damage Assessment Using Mosaic Imagery from Hurricane Milton [link]

2025 Texas A&M Institute of Data Science (TAMIDS), TAMIDS Announces 2025-2026 Cohort of Student Ambassadors [link]

2024 Texas A&M Engineering News, Throwing Shade at Heatwaves [link]

2024 Esri Industry Blog, Showcasing Innovation in GIS Education through Student Projects at the LA Geospatial Summit [link]